

Abhigyan Gandhi

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PROFESSIONAL SUMMARY

Graphics and games engineer specialising in real-time rendering and GPU optimization. MSc graduate from the University of Leeds; recipient of the Game Republic Red Kite Games Technology Award 2025. Proficient in C++, C#, Python, GitHub, VS Code, Graphics APIs and game engines with experience shipping production-quality renderers.

SKILLS

- **Programming:** C++, C#, GLSL, HLSL, CUDA, Python, TypeScript, SQL.
- **Graphics and Game Dev:** Vulkan, OpenGL, DirectX12, SPIR-V, Unity, Unreal Engine, Physics Simulation, Game Engine Architecture, HPC/GPU Programming, ECS architecture, SIMD, memory management, networking, PBR, BRDF, shadows, GPU descriptors, synchronization primitives, Ray Tracing, Descriptor indexing, OpenCL, Render Graph.
- **Tools:** Git, Linux, GitLab CI/CD, Docker, Node.js, MATLAB, Azure DevOps, RenderDoc, Tracy, Nsight, MySQL, Bash.
- **AI/ML:** TensorFlow, Scikit-Learn, Pandas, CNNs, GANs, Computer Vision.
- **Languages:** English(Native), Hindi(Native).
- **Technical highlights:** Shader programming, GPU optimization and profiling, physics simulation, real-time rendering, ECS architecture and gameplay pipelines.

PROJECTS

- **Game Engine Development (C++, Vulkan, 2025):** Developed multi-pass Vulkan renderer with PBR pipeline; implemented GPU culling, ECS and vehicular physics simulation; achieved **100+ FPS** on RTX hardware; won **Game Republic Red Kite Games Game Technology Award** at the [Student Showcase 2025](#) (Group project). [\[GitHub Repo\]](#)
- **Vulkan Ray Tracer (C++, Vulkan, 2026):** Built a hardware-accelerated ray tracer using **VK_KHR_ray_tracing_pipeline**; implemented BLAS/TLAS, Shader Binding Table, and **recursive ray tracing (shadows, reflections)**. [\[GitHub Repo\]](#)
- **LETTER SHIFT (Unity, C#, 2025):** Developing a daily puzzle game for mobile and web; preparing for release on [letter-shift.com](#) and Android (APK available on portfolio). [\[GitHub Repo\]](#)
- **Ray Tracing and Rasterization (C++, OpenGL, Vulkan, 2024):** Implemented CPU-based ray tracer and GPU rasterizer to compare performance and visual fidelity. Reduced GPU render pass time by **35%** through descriptor sets.
- **Bezier Curves and Physics Simulation (C++, OpenGL, 2024):** Created a Bezier curve renderer and physics-based animation with collision detection for a character and bouncing dodecahedra.

WORK EXPERIENCE

Tech Designer (Freelance) | Creative AF Games | Mar 2026 - Present

- Engineered character ability systems and gameplay mechanics in **Unreal Engine 5** using **Blueprints** and **C++**.
- Optimised runtime performance identifying and resolving **CPU/GPU bottlenecks** to maintain frame budget targets.
- Collaborated with design and engineering teams in an **agile pipeline** to implement and iterate on technical features.

Ticket Sales Advisor (Part-time) | Leeds United | Oct 2025 - Present

- Resolved **time sensitive problems efficiently** in a fast-paced, high pressure environment, ensuring a positive exp.
- Delivered **calm, professional customer service**, assisting supporters with ticketing issues, collections and enquiries.

Game Development Intern | Sentient Labs | Jun 2020 - Aug 2020

- Developed a **WebGL** game in **Unity**; lifted user engagement by **25%** and boosted average session duration by **18%**.
- Implemented responsive gameplay mechanics and modular UI systems, reducing UI load time by **40%**.
- Integrated **50+ art and audio assets**, ensuring visual consistency at a sustained **60 FPS**.
- Collaborated in agile sprints, completing **100%** of assigned tasks using Git-based version control.

AI/ML Intern | Stella Stays | Aug 2021 - Jan 2022

- Implemented predictive pricing model using **LSTM** and **regression**, improving base price accuracy to **10% MAPE**.
- Built a NLTK-based sentiment analysis pipeline for hotel reviews, achieving **80% accuracy**.
- Developed a room allocation algorithm based on **minimum slack** and **clustering**; led to a **+10% optimized** allocator.

EDUCATION

MSc. High-Performance Graphics and Games Engineering, Merit | University of Leeds | Sept 2024 - Sept 2025

- **Modules:** Foundation of Computer Graphics, Advanced Rendering, Modelling and Animation.
- **Dissertation:** Evaluating the performance and visual fidelity of INTEL OIDN against NVIDIA NRD. (Key finding: Hybrid system including real-time and offline denoising is recommended for best performance)

B.E. (Hons) Computer Science, Data Science, CGPA: 8.72 | BITS PILANI DUBAI CAMPUS | Sept 2018 - Sept 2022

- **Modules:** Machine Learning, Data Mining, Foundations of Data Science, Neural Networks.
- **Final Year Design Project:** Mobile Robot Path Planning using Deep Learning Techniques.